*2. A gas bubble rises from the bottom of a lake to the water surface. Its radius increases from 0.8 cm to 1.0 cm.

(a) If the gas pressure in the bubble at the water surface is 1.01×10^5 Pa, find the gas pressure in the bubble when it is at the bottom of the lake. Assume that the temperature of the gas in the bubble remains constant. (2 marks)

(b) Use kinetic theory to explain the change in gas pressure in the bubble as its volume increases. (2 marks)

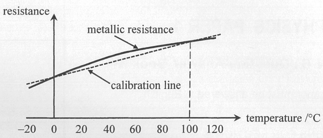
Answers written in the margins will not be marked.

2015 Q2

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. The solid curve in Figure 1.1 shows how the resistance of a metallic resistance thermometer varies with temperature. This thermometer is calibrated at standard atmospheric pressure for the melting point of ice and the steam point of boiling water. The dotted calibration line represents how the resistance of the thermometer varies with temperature if a linear resistance-temperature relationship is assumed. The deviation of the curve from linearity has been exaggerated in the figure.

Figure 1.1



(a) (i) Using the resistances at the calibration points tabulated below, calculate the expected resistance at 60 °C if the resistance varies linearly with temperature. (2 marks)

temperature /°C	**resistance /Ω**
0	102.00
100	140.51

(ii) Now if the resistance of the resistance thermometer is the value found in (a)(i), is the actual temperature higher than, lower than or equal to 60 °C ? (1 mark)

(b) In an experiment to determine the specific heat capacity of water c_W, Peter used this calibrated resistance thermometer to measure the temperature of water being heated from 0 °C to 60 °C. Heating was stopped when this thermometer's resistance reached the value found in (a)(i). Assuming negligible heat exchange with the surroundings, no error in measuring the energy supplied and the mass of water, explain whether the experimental value of c_W found is higher than, lower than or the same as the actual value. (2 marks)

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*2. The aqua-lung (a cylinder containing compressed air) for divers has a capacity of 1.0×10^4 cm^3. When the aqua-lung is filled, the air inside has a pressure of 210 atm (atmospheric pressure) at 24 °C. The air in the aqua-lung is allowed to expand through a pressure-reducing valve until its pressure equals that of the surrounding water before it is supplied to divers. Assume that the temperature of the air inside the aqua-lung is always equal to that of the surrounding water.

(a) A diver stays in water of temperature 24 °C and pressure 2.0 atm at a depth of 10 m. Find the total volume of air (in cm^3) available for the diver from the aqua-lung at this water pressure. (2 marks)

(b) The supply of air in (a) is sufficient for the diver to remain at such a depth for 1 hour.

(i) If the diver breathed in the same volume V_0 (in cm^3) of air per minute, find V_0. (1 mark)

(ii) If the diver dives deeper where the water is of temperature 20 °C and pressure 4.5 atm, estimate how long (in minutes) the air in a fully-filled aqua-lung would last. Assume that the diver breathes in the **same volume of air** per minute as that found in (b)(i). (3 marks)

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mass of gas at constant temperature. The volume V of the gas trapped is read directly from the syringe and the corresponding pressure p is measured by a data-logger via a pressure sensor.

(a) The initial volume and pressure of the gas are 6.0×10^{-5} m^3 and 1.0×10^5 Pa respectively at a room temperature of 25 °C. Estimate the number of gas molecules trapped in the syringe. (3 marks)

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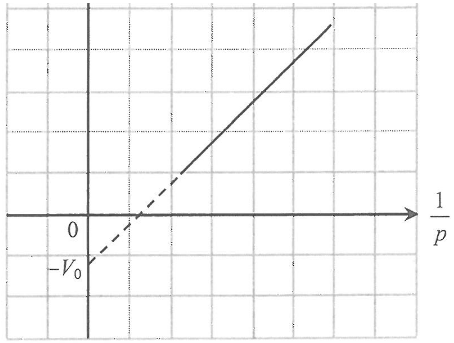

Figure 2.2

(i) State **ONE** experimental precaution for keeping the gas temperature constant. (1 mark)

(ii) The straight line graph does not pass through the origin but cuts the vertical axis at $-V_0$ instead. Suggest what V_0 stands for. (1 mark)

(iii) If the experiment is repeated at a higher room temperature using this set-up with the same mass of the same gas, sketch the expected graph in Figure 2.2. (2 marks)

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*3. The average kinetic energy of one monatomic gas molecule at temperature T is given by

$$E_K = \frac{3}{2}\left(\frac{R}{N_A}\right)T,$$

where R is the universal gas constant and N_A is the Avogadro constant. A monatomic gas is heated from 300 K to 350 K under fixed volume.

(a) Estimate the ratio of the root-mean-square speed ($c_{r.m.s.}$) of the gas molecules at the two temperatures ($\frac{c_{r.m.s.} \text{ at } 350\,\text{K}}{c_{r.m.s.} \text{ at } 300\,\text{K}}$). (2 marks)

(b) Thus, using kinetic theory, explain why the gas pressure would increase. (2 marks)

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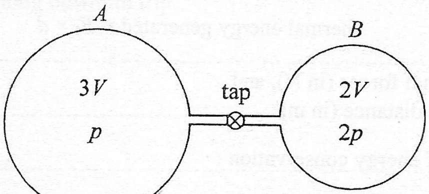
*2. Two vessels, A and B, of volumes $3V$ and $2V$ respectively are connected by a narrow tube with a tap as shown in Figure 2.1. Initially the tap is closed and both vessels are at the same temperature. Vessel A contains helium gas at a pressure p while vessel B contains 0.8 mol helium gas at a pressure $2p$. Assume that helium gas can be taken as an ideal gas.

Figure 2.1



(a) Deduce the amount of helium gas (in mol) in vessel A. (2 marks)

(b) Now the tap is open and a steady state is reached. Assume that the temperature remains unchanged.

(i) Find the gas pressure inside the vessels in terms of p. (2 marks)

(ii) Account for the pressure change of the gas in vessel A using kinetic theory. (2 marks)

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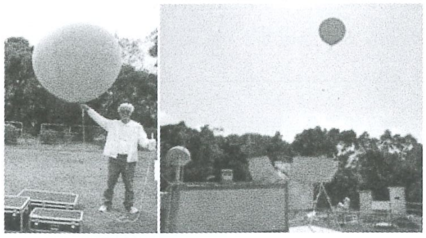
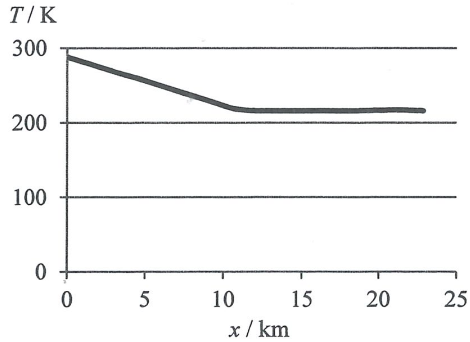
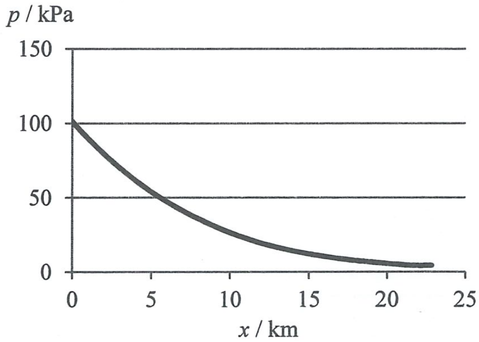
*2. A weather balloon of volume 0.52 m^3 is filled with helium gas of temperature 15 °C and pressure 100 kPa at ground level.



(a) Find the amount of helium gas (in mol) in the balloon. (2 marks)

(b) The following graphs show the variation of air temperature T and atmospheric pressure p with height x above ground level.





The weather balloon is released and rises to the upper atmosphere. Assume that the temperature and pressure of the helium gas in the balloon are the same as those of the air outside at any height x.

(i) A student believes that as the air temperature decreases in the first 10 km, the volume of the balloon decreases. Referring to the graphs above, explain qualitatively why this belief is not correct. (2 marks)

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(ii) In fact the weather balloon keeps on expanding when it rises. The air temperature becomes steady at 216 K from a height of 12 km onwards. When the balloon rises further beyond 12 km and its volume reaches 8 m^3,

(1) estimate the gas pressure in the balloon; (2 marks)

(2) hence find the corresponding height reached by the balloon. The variation of atmospheric pressure p with height x (in km) is given by

$$p = p_0 e^{-kx},$$

where p_0 is the atmospheric pressure at ground level and $k = 0.138\ km^{-1}$. (2 marks)

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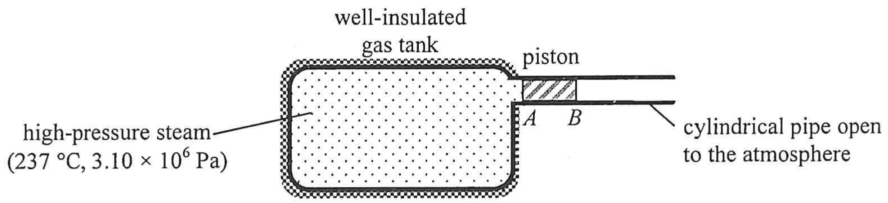
2. Figure 2.1 shows a large gas tank connected with a cylindrical pipe open to the atmosphere. The pipe is fitted with a smooth piston AB. This well-insulated gas tank is filled with high-pressure steam at a temperature of 237 °C under a pressure of 3.10×10^6 Pa while the movable piston is held stationary by a force F_p.
Given: atmospheric pressure = 1.0×10^5 Pa

Figure 2.1



(a) (i) On Figure 2.1 indicate the force F_p. (1 mark)

*(ii) By considering the force acting on the piston due to the difference in pressure, find the value of F_p. The piston has a cross-sectional area of 0.67 m^2. (2 marks)

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*(iii) Estimate the volume of the gas tank which contains 570 kg steam. You may treat the steam as an ideal gas. Given: mass of one mole of steam = 0.018 kg. (3 marks)

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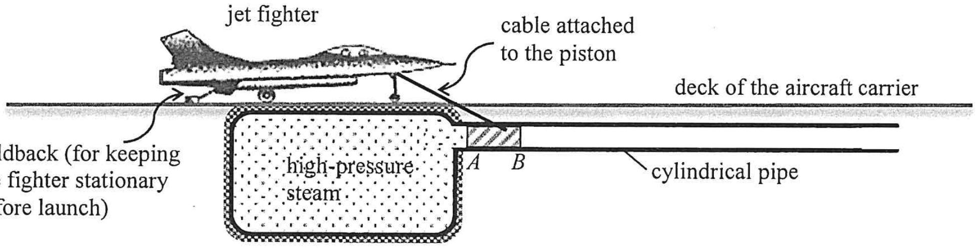
(b) This set-up can be used as a 'steam catapult' to launch jet fighters from an aircraft carrier. A jet fighter in position to be launched is connected to the piston via an inextensible cable as shown in Figure 2.2. When the holdback behind the jet fighter is released, the high-pressure steam in the gas tank expands and pushes the piston which in turn helps to accelerate the jet fighter.

Figure 2.2



In a trial run of the catapult, a jet fighter (with its engine shut down) acquires a final speed of 54 m s^{-1} in 1.5 s after running a distance horizontally on the deck. The mass of the jet fighter is 2.6×10^4 kg.

(i) Find the work done by the net force on the jet fighter during launch. (2 marks)

(ii) Calculate the average acceleration of the jet fighter during launch. (2 marks)

*(iii) State whether the acceleration of the jet fighter is increasing, decreasing or uniform during launch. Explain your answer. (3 marks)

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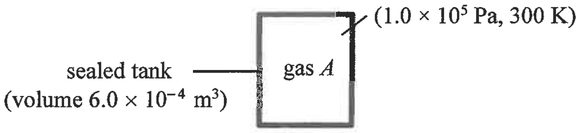
*2. (a) (i) Figure 2.1 shows a sealed tank of volume 6.0×10^{-4} m^3 containing a monatomic gas A at a pressure of 1.0×10^5 Pa and at a temperature of 300 K.

Figure 2.1

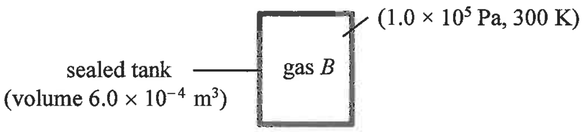


(I) Estimate the number of gas molecules in the tank, N. (2 marks)

(II) Estimate the average kinetic energy of the gas molecules, E_K. (2 marks)

(ii) Figure 2.2 shows another identical tank containing monatomic gas B under the same pressure and temperature. A molecule of gas B has $\frac{1}{5}$ the mass of a molecule of gas A.

Figure 2.2



(I) State whether N and E_K of gas B are larger than, smaller than or the same as the corresponding values of those of gas A found in (a)(i). (2 marks)

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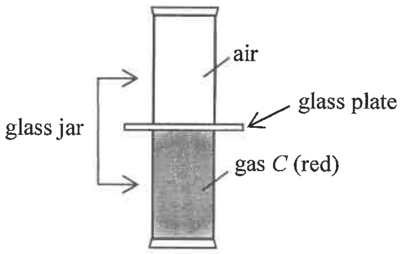
(II) Given that the root-mean-square speed ($c_{r.m.s.}$) of the molecules of gas A is 600 m s^{-1}, estimate $c_{r.m.s.}$ of the molecules of gas B. (2 marks)

(b) Figure 2.3 shows two glass jars containing air and gas C respectively and they are separated by a glass plate. Both are at the same pressure and temperature. Gas C is red in colour.

Figure 2.3



After removing the glass plate, it takes a few minutes for gas C to diffuse several centimetres into the upper jar despite its molecules having a root-mean-square speed of 200 m s^{-1}. Explain this observation. (2 marks)

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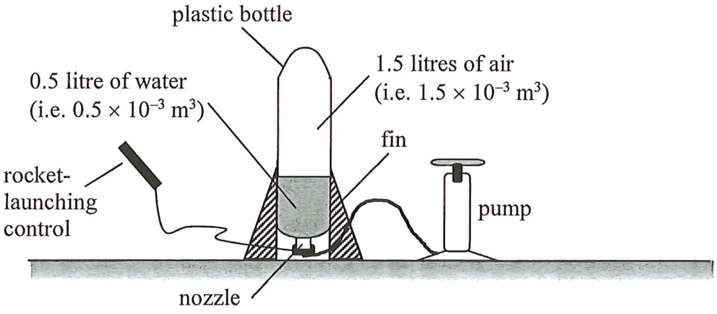
*2. Figure 2.1 shows a 'water rocket' built from a 2-litre plastic bottle with fins attached. The bottle is filled with water such that 1.5 litres of air is trapped inside.

Figure 2.1



Initially the trapped air of temperature 27 °C is at atmospheric pressure.
Given: atmospheric pressure = 1.0×10^5 Pa
density of air = 1.20 kg m^{-3}

(a) Find the mass of the trapped air m_0. (2 marks)

Air is then pumped slowly into the rocket until the total mass of trapped air inside the bottle becomes $4m_0$. Assume that the trapped air is kept at a temperature of 27°C (i.e. 300 K) and its volume remains constant.

(b) (i) Find the pressure of the trapped air. (2 marks)

(ii) Use kinetic theory to explain why the pressure of the trapped air increases. (2 marks)

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(iii) Given that the area of the water surface is 0.014 m^2. Estimate the **increase** in the force acting on the water due to the trapped air. (2 marks)

(c) Given that the mass of the plastic bottle is negligible.

(i) After the rocket-launching control is pulled and a jet of water ejects from the nozzle, explain why the rocket can go upwards using Newton's laws of motion. (2 marks)

(ii) **Just** after pulling the control, what is **most** of the energy stored in the pressurized air converted to ? (1 mark)

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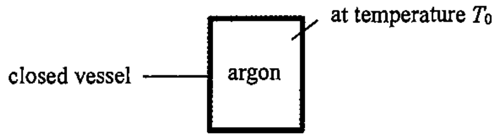
*2. Figure 2.1 shows a closed vessel of a fixed volume containing argon, a monatomic gas, maintained at temperature T_0. The root-mean-square speed ($c_{r.m.s.}$) of the argon atoms inside is 500 m s^{-1}.

Figure 2.1



(a) (i) Calculate the average kinetic energy of the argon atoms, E_K. Given that the mass of an argon atom is 6.63×10^{-26} kg. (2 marks)

(ii) Hence, or otherwise, find T_0 (in K). (2 marks)

(b) If some of the argon has been leaked from the vessel while the temperature remains at T_0, state and explain the change, if any, in the root-mean-square speed of the argon atoms inside the vessel. (2 marks)

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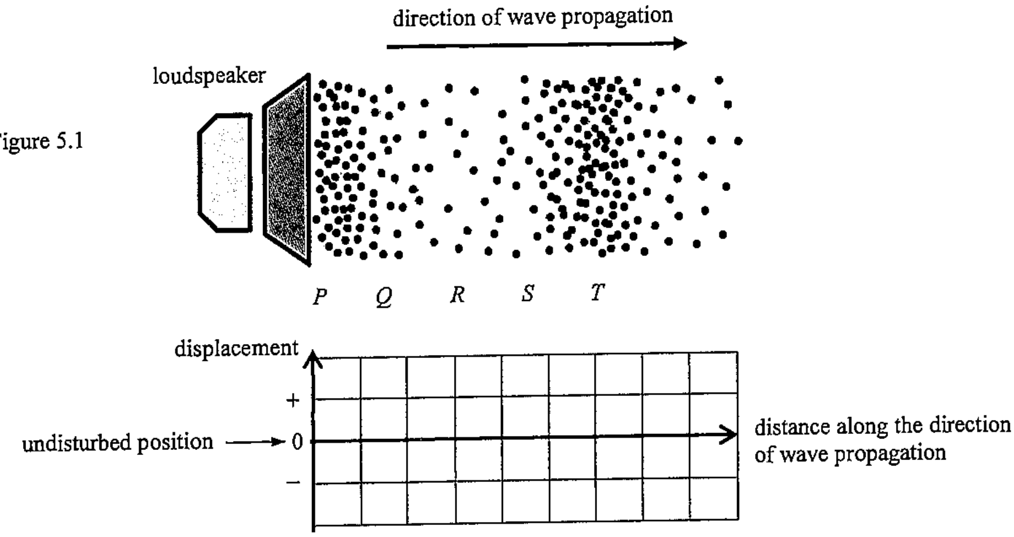
5. Figure 5.1 represents the distribution of air particles at a certain instant in a sound wave produced by a loudspeaker.

Figure 5.1



(a) Sketch the displacement-distance graph of the air particles at that instant in the grid provided. Take the displacement to the right as positive (+). (2 marks)

(b) At the instant shown, state a region (*P*, *Q*, *R*, *S* or *T*) in which

(i) the air particles are moving with the highest speed; (1 mark)

(ii) the air pressure is the lowest. (1 mark)

9. An ideal gas is at temperature 0 °C and pressure 1.01×10^5 N m^{-2}.

(a) *(i) Find the volume of one mole of the gas. (2 marks)

(ii) Find the number of molecules in 1 m^3 of the gas under the conditions given. (2 marks)

*(b) At a certain location above the Earth in space where the pressure is 2.75×10^{-10} N m^{-2}, 1 m^3 of the gas contains about 1.00×10^{11} molecules. Assuming the gas is ideal, estimate the temperature at the location (in K). (2 marks)

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(b) The power output of the heater is 2 kW. Estimate the time required. (2 marks)

(c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on. (1 mark)

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(b) (i) Find the average kinetic energy of the gas molecules. (2 marks)

(ii) The mass of one mole of neon gas is 0.020 kg. Find the root-mean-square speed of the gas molecules. (2 marks)

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(1 mark)

(b) Estimate the maximum magnetic flux linkage produced by the wire through the search coil. (3 marks)

(c) A student claims that the search coil can be used to measure the resultant magnetic field produced by the Earth and the wire. Explain briefly whether this claim is correct. (2 marks)

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Answer **ALL** questions. Parts marked with "*" involve knowledge of the extension component. Write your answers in the spaces provided.

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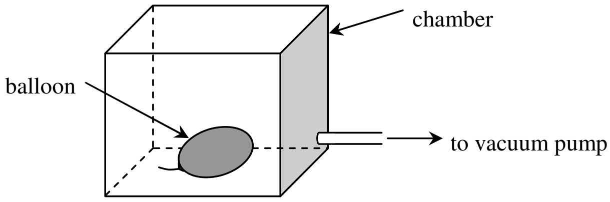

Figure 1.1

A balloon containing 0.01 m^3 of gas at a pressure of 100 kPa is placed inside a chamber. Air is slowly pumped out from the chamber while the temperature remains unchanged.

*(a) Explain, in terms of molecular motion, how the gas inside the balloon exerts a pressure on its inner surface. (2 marks)

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*(b) Find the final pressure inside the balloon when its volume is doubled. (2 marks)

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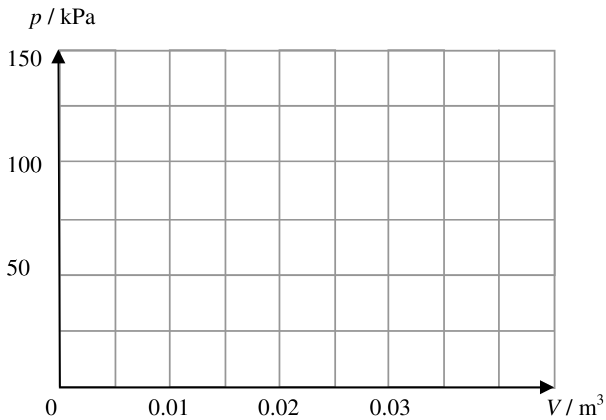
*(c) Sketch a graph to show the relationship between the pressure p inside the balloon and the volume V of the balloon. (2 marks)



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